MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§1.4

The Babylonian Numeration System

Two symbols, fifty-nine digits, and the first place value system

Today's Goal

After completing this section, you should be able to:

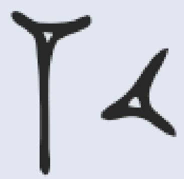
- Read and write numbers using the Babylonian numeration system.

LT.2 Numeration Systems

I can convert numbers between various numeration systems including Roman, Egyptian, Mayan, Babylonian, and Hindu-Arabic.

How Many Symbols Would You Need?

The Egyptians used seven symbols. The Romans used seven. How many do you think the Babylonians needed?



Two.

Only Two Symbols

Table 1.4.2. Babylonian Symbols

Babylonian Numeral	Number
𒁹	1
𒌋	10

Every Babylonian numeral is built from these two marks: the wedge and the corner hook.

Counting by Sixties

The first known place value system came from the Sumerians of Mesopotamia around 3000 BC, and was later refined by the Babylonians. Unlike our own system, it is **base-60**. That means two things.

1. Each position is worth a power of 60, not a power of 10.
2. They needed to write the numbers from 1 to 59. Those are their “digits”.

Table 1.4.1. Base-60 Place Values

Position	n	$\ldots$	4th	3rd	2nd	1st
Place Value	60^{n-1}	$\ldots$	$60^3 = 216{,}000$	$60^2 = 3{,}600$	60	1

Writing 1 Through 9

Stack the wedge to count.

Table 1.4.3. Babylonian Numerals (Ones)

Babylonian Numeral	𒁹	𒈫	𒐈	𒐉	𒐊	𒐋	𒐌	𒐍	𒐎
Number	1	2	3	4	5	6	7	8	9

Writing 10 Through 50

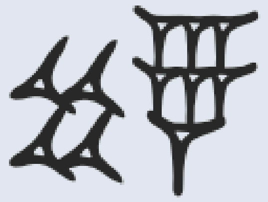
Stack the corner hook the same way.

Table 1.4.4. Babylonian Numerals (Tens)

Babylonian Numeral	𒌋	𒎙	𒌍	𒐏	𒐐
Number	10	20	30	40	50

How Would You Write 47?

You have the corner hook worth 10 and the wedge worth 1. Put them together to make 47.



Four corner hooks, then seven wedges. Tens first, then ones.

One Symbol Set, Fifty-Nine Digits

Table 1.4.5. Babylonian Numerals (1 to 59)

𒁹 = 1	𒌋𒁹 = 11	𒎙𒁹 = 21	𒌍𒁹 = 31	𒐏𒁹 = 41	𒐐𒁹 = 51
𒈫 = 2	𒌋𒈫 = 12	𒎙𒈫 = 22	𒌍𒈫 = 32	𒐏𒈫 = 42	𒐐𒈫 = 52
𒐈 = 3	𒌋𒐈 = 13	𒎙𒐈 = 23	𒌍𒐈 = 33	𒐏𒐈 = 43	𒐐𒐈 = 53
𒐉 = 4	𒌋𒐉 = 14	𒎙𒐉 = 24	𒌍𒐉 = 34	𒐏𒐉 = 44	𒐐𒐉 = 54
𒐊 = 5	𒌋𒐊 = 15	𒎙𒐊 = 25	𒌍𒐊 = 35	𒐏𒐊 = 45	𒐐𒐊 = 55
𒐋 = 6	𒌋𒐋 = 16	𒎙𒐋 = 26	𒌍𒐋 = 36	𒐏𒐋 = 46	𒐐𒐋 = 56
𒐌 = 7	𒌋𒐌 = 17	𒎙𒐌 = 27	𒌍𒐌 = 37	𒐏𒐌 = 47	𒐐𒐌 = 57
𒐍 = 8	𒌋𒐍 = 18	𒎙𒐍 = 28	𒌍𒐍 = 38	𒐏𒐍 = 48	𒐐𒐍 = 58
𒐎 = 9	𒌋𒐎 = 19	𒎙𒐎 = 29	𒌍𒐎 = 39	𒐏𒐎 = 49	𒐐𒐎 = 59
𒌋 = 10	𒎙 = 20	𒌍 = 30	𒐏 = 40	𒐐 = 50	

Reading a Babylonian Numeral

Procedure 1.4.6. Converting a Babylonian Numeral to the Corresponding Hindu-Arabic Numeral.

To convert a Babylonian numeral to the corresponding Hindu-Arabic numeral, do the following.

1. Determine the number of digits in the numeral.
2. Identify the place value of each digit in the numeral.
3. Multiply each digit by its corresponding place value.
4. Add the results from the previous step.

Reading a Two-Digit Numeral

What number is represented by this Babylonian numeral?

𒐈 𒌍𒈫

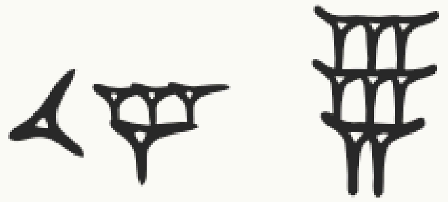
Two digits, separated by a space.

Position	Place Value	Digit	Value of Digit
1st	1	𒌍𒈫 $= 32$	$32 \cdot 1 = 32$
2nd	60	𒐈 $= 3$	$3 \cdot 60 = 180$

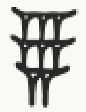
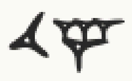
Adding gives $180 + 32 = 212$, the Hindu-Arabic numeral 212.

Reading Another Two-Digit Numeral

What number is represented by this Babylonian numeral?



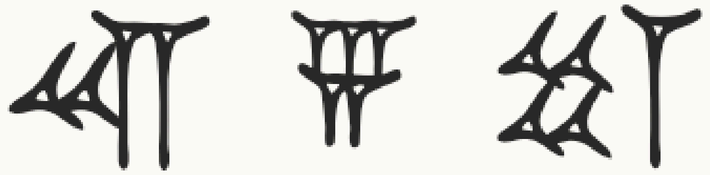
Again two digits, separated by a space.

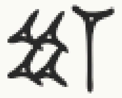
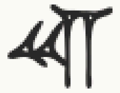
Position	Place Value	Digit	Value of Digit
1st	1	 $= 8$	$8 \cdot 1 = 8$
2nd	60	 $= 14$	$14 \cdot 60 = 840$

Adding gives $840 + 8 = 848$, the Hindu-Arabic numeral 848.

Reading a Three-Digit Numeral

What number is represented by this Babylonian numeral?



Position	Place Value	Digit	Value of Digit
1st	1	 $= 41$	$41 \cdot 1 = 41$
2nd	60	 $= 5$	$5 \cdot 60 = 300$
3rd	$60^2 = 3{,}600$	 $= 22$	$22 \cdot 3{,}600 = 79{,}200$

Adding gives $79{,}200 + 300 + 41 = 79{,}541$, the Hindu-Arabic numeral $79{,}541$.

Now You Try: Reading

You Try 1.4.1. What number is represented by this Babylonian numeral?

𒌋𒁹 𒌍𒐈 𒐎

Position	Place Value	Digit	Value of Digit
1st	1	𒐎 $= 9$	$9 \cdot 1 = 9$
2nd	60	𒌍𒐈 $= 33$	$33 \cdot 60 = 1{,}980$
3rd	$60^2 = 3{,}600$	𒌋𒁹 $= 11$	$11 \cdot 3{,}600 = 39{,}600$

Total. $39{,}600 + 1{,}980 + 9 = 41{,}589.$

Writing a Babylonian Numeral

Procedure 1.4.8. Converting a Hindu-Arabic Numeral to the Corresponding Babylonian Numeral.

To convert a Hindu-Arabic numeral to the corresponding Babylonian numeral, do the following.

1. If the number is less than 60, then write the corresponding Babylonian numeral using Table 1.4.5 and we are done. Otherwise, go to the next step.

Writing a Numeral, continued

Procedure 1.4.8, continued.

2. Determine the highest base-60 place value needed for the number. Divide the number by this place value.
   a. Use Table 1.4.5 to write the quotient as a Babylonian numeral in the corresponding place value.
   b. Return to the first step and use the remainder of this division instead of the original number.

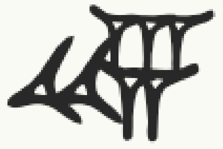
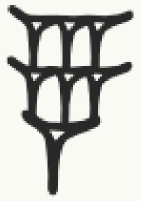
Writing a Number Under 3,600

Write the number $1{,}507$ as a Babylonian numeral.

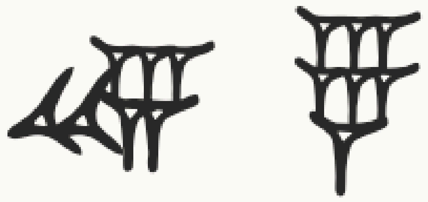
Since $60 < 1{,}507 < 60^2$, we will need two digits. So we divide $1{,}507$ by 60.

$$1{,}507 \div 60 = 25 \text{ remainder } 7$$

The quotient is 25 and the remainder is 7. We write each as a Babylonian numeral.

$25 =$  $7 =$ 

These are the two digits in our number, so:

$1{,}507 =$ 

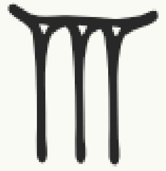
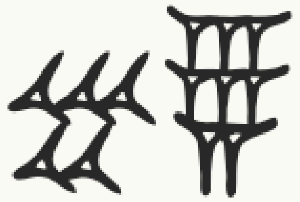
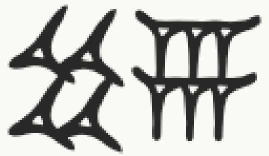
Writing a Number Over 3,600

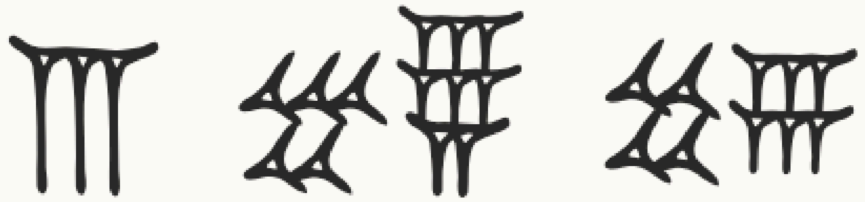
Write the number $14{,}326$ as a Babylonian numeral.

Since $60^2 < 14{,}326 < 60^3$, we need three digits.

$$14{,}326 \div 3{,}600 = 3 \text{ remainder } 3{,}526$$

$$3{,}526 \div 60 = 58 \text{ remainder } 46$$

$3 =$  $58 =$  $46 =$ 

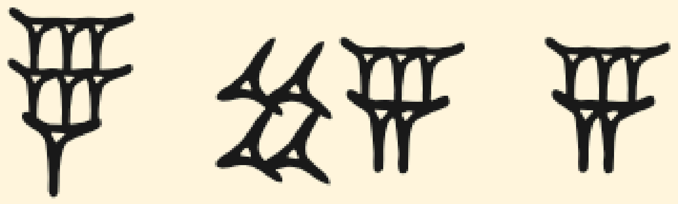
$14{,}326 =$ 

Now You Try: Writing

You Try 1.4.2. Write the number $27{,}905$ as a Babylonian numeral.

$$27{,}905 \div 3{,}600 = 7 \text{ remainder } 2{,}705$$

$$2{,}705 \div 60 = 45 \text{ remainder } 5$$

$$27{,}905 =$$ 

The Spacing Can Fool You

Warning 1.4.11

These three numerals are not the same. Only the spacing differs.

𒐖 𒌋𒐗 $= 133$

𒐖   𒌋𒐗 $= 7{,}213$

𒐖   𒌋𒐗 $= 432{,}013$

Counting the Positions

- The first is a **two-digit** numeral.
- The second is a **three-digit** numeral with nothing in the second position.
- The third is a **four-digit** numeral with nothing in the second and third positions.

A wider gap means an empty position. That is easy to see when the three sit side by side, and easy to miss otherwise.

A Mark for an Empty Position

Around 400 BC the Babylonians introduced a placeholder symbol — two slanted wedges — to mark an empty position.

Since $7{,}213 = 2 \times 60^2 + 0 \times 60 + 13$ has nothing in the 60s position, that position gets the placeholder.

 $= 7{,}213$

Now this numeral can no longer be confused with 133.

Two Marks, Sixty Places

- Two symbols only: the wedge 𒁹 is 1, the corner hook 𒌋 is 10.
- They combine into 59 digits, and each position is worth a power of 60.
- **Reading:** multiply each digit by its place value, then add.
- **Writing:** divide by the highest place value, then keep dividing each remainder by the next one down.
- Watch the spacing — a wider gap means an empty position.